

LESSON 01 / 第一課

Compartmentalization of Decay in Trees: From Stand Patterns to Cell Walls

樹木腐朽區隔化：從林分格局到細胞壁

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Trees do not heal wood in the animal sense. They restrict the spread of decay and dysfunction after wounding, then continue growth outside the damaged volume. That process — compartmentalization — is hierarchical. At stand scale, species differ in how far a decay column travels after a comparable wound. At tree scale, Shigo distinguished a reaction zone laid down in wood that was already present at injury from a barrier zone laid down by the cambium after injury. At tissue scale those become Walls 1–3 and Wall 4. At cell scale the same story is suberin, phenolics, tyloses and pit aspiration. The four walls are not equally strong, and a successful barrier is not a mechanical-safety certificate. This review keeps those two results in view while moving from published species patterns to the chemistry that those patterns imply.

樹木不以動物的意義癒合木材。它們在受傷後限制腐朽與功能障礙的擴散，然後在受損體積之外繼續生長。該過程——區隔化——是層級的。林分尺度上，樹種在可比傷口之後，腐朽柱能走多遠並不相同。整株尺度上，Shigo 區分受傷時已存在木材中的反應區，與受傷後形成層新造的屏障區。組織尺度上它們成為第 1–3 壁與第 4 壁。細胞尺度則是木栓質、酚類、侵填體與紋孔封閉。四壁並非同等強，屏障成功也不是力學安全證明。本綜述在由已發表樹種格局寫到其所蘊含化學的同時，把這兩項結果留在視野內。



Figure 1. Nested scales used to organise this review, from stand patterns to cell-wall chemistry.

圖 1. 本綜述的嵌套尺度：由林分格局到細胞壁化學。

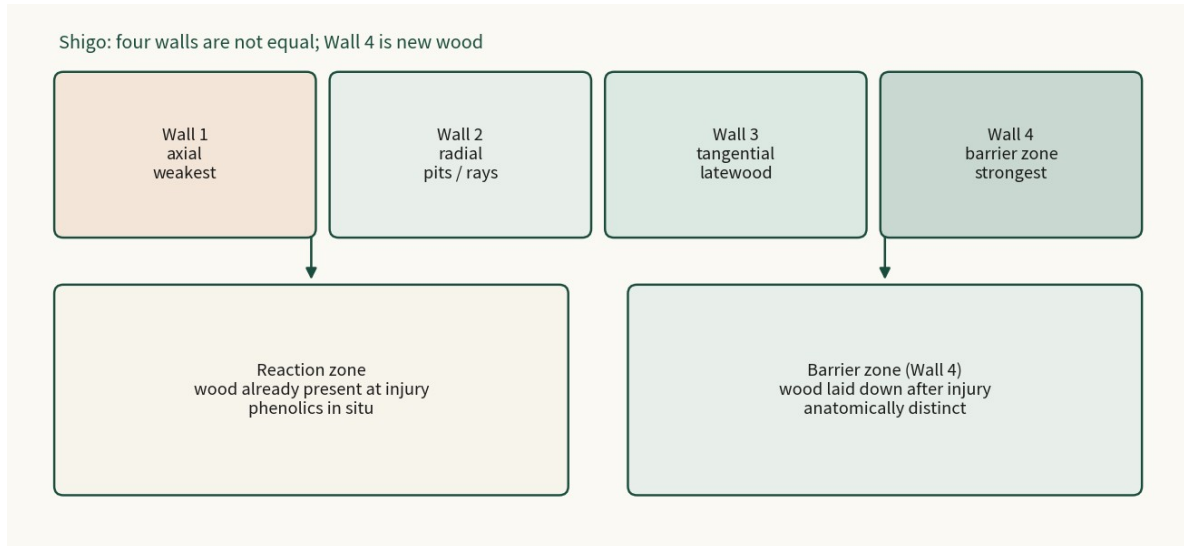


Figure 2. Reaction zone (wood present at injury) versus barrier zone (wood formed after injury). Walls 1–3 are not Wall 4.
圖2. 反應區（受傷時已有的木材）對屏障區（受傷後新造的木材）。第1–3壁不是第4壁。

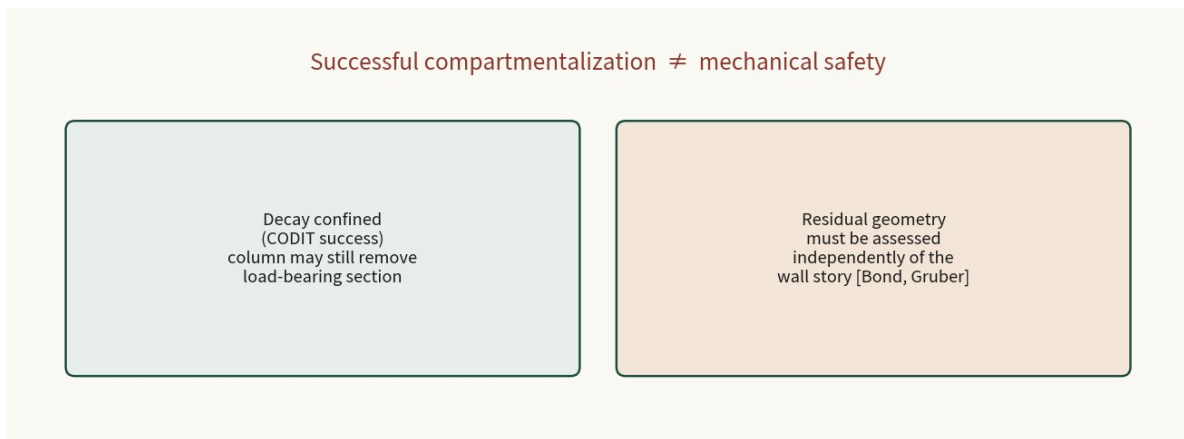


Figure 3. Compartmentalization success and residual load-bearing geometry are separate questions.
圖3. 區隔化成功與剩餘承載幾何是分開的問題。

1. Scope of the review

1. 綜述範圍

The object is how woody plants restrict decay and dysfunction after wounding or infection. The organising principle is scale (Fig. 1). Section 2 treats species and stand patterns. Section 3 treats Shigo's reaction-zone / barrier-zone distinction. Section 4 treats the four walls as tissue anatomy. Section 5 treats cell-wall chemistry. Section 6 records what compartmentalization does not decide.

對象是木本植物在受傷或感染後如何限制腐朽與功能障礙。組織原則是尺度（圖1）。第2節處理樹種與林分格局。第3節處理 Shigo 的反應區／屏障區區分。第4節把四壁當成組織解剖。第5節處理細胞壁化學。第6節記錄區隔化所不能決定的事。

2. Stand and species patterns

2. 林分與樹種格局

Shigo and Marx, and Shigo's later syntheses, generalised from decades of dissection that trees wall off damaged tissue rather than restore it [1,2]. Comparative work since then has shown large species effects. Many ring-porous temperate angiosperms confine axial spread more effectively than some diffuse-porous

taxa; gymnosperms rely more on constitutive resin and phenolics than on a classic four-wall anatomy [3,4]. Schwarze documented that white-rot, brown-rot and soft-rot fungi breach reaction zones by different enzymatic and hyphal strategies, so ‘the fungus’ is not a single opponent [5].

Shigo 與 Marx，以及 Shigo 其後的綜述，由數十年解剖概括：樹木把受損組織封起來，而不是把它修復 [1,2]。其後比較工作顯示巨大的樹種效應。許多環孔的溫帶被子植物，軸向限制優於部分散孔類群；裸子植物較依賴構成性樹脂與酚類，而不是經典四壁解剖 [3,4]。Schwarze 記錄白腐、褐腐與軟腐真菌以不同酶與菌絲策略突破反應區，因此「真菌」不是單一對手 [5]。

Those stand-scale differences are the first result that must survive translation into practice. A wound of a given diameter is not a constant event across taxa. The same pruning cut that a strong compartmentalizer confines to a short column can, in a weak compartmentalizer, become a long decay pipe. Species tables that ignore this are not physiology; they are planting lists.

這些林分尺度差異，是翻譯到實務時必須存活的第一項結果。給定直徑的傷口，在不同分類群上不是常數事件。強區隔者能把同一修剪切面封成短柱；弱區隔者則可能變成一條長腐朽管。忽略這一點的樹種表不是生理學，只是種植名單。

3. Tree scale: reaction zone and barrier zone

3. 整株尺度：反應區與屏障區

Shigo’s durable distinction is temporal [1,2]. The reaction zone is a chemical modification of wood that already existed when the injury occurred. The barrier zone is anatomically new wood laid down by the cambium after the injury (Fig. 2). Calling both of them ‘walls’ without that clock collapses two different tissues into one diagram. Dujesiefken and Liese’s CODIT-principle restatement keeps the same clock and emphasises that the barrier zone is the strongest of the four walls precisely because it is new wood [6].

Shigo 可持久的區分是時間上的 [1,2]。反應區是對受傷時已存在木材的化學修飾。屏障區是受傷後形成層新造、解剖上不同的木材（圖 2）。若把兩者都叫「壁」而不帶此時鐘，就把兩種組織壓成一張圖。Dujesiefken 與 Liese 的 CODIT 原則重述保留同一時鐘，並強調屏障區之所以是四壁中最強，正因為它是新木材 [6]。

Pearce reviewed the reaction zone as a dynamic, metabolically expensive front rather than a static stain [7]. The visible colour change is not itself the defence; it is a correlate of phenolics and related deposits. A reaction zone can be breached and then re-formed. Treating a stain boundary on a cut face as a finished verdict is therefore a misreading of the tissue.

Pearce 把反應區綜述為動態、代謝昂貴的鋒面，而不是靜態色斑 [7]。可見的顏色變化本身不是防禦，而是酚類及相關沉積的相關物。反應區可被突破然後再形成。因此把切面上的色斑邊界當成完成判決，是對組織的誤讀。

4. Tissue scale: Walls 1–4

4. 組織尺度：第 1–4 壁

Wall 1 resists axial spread and is, in Shigo’s ranking, the weakest. Wall 2 resists inward radial spread (latewood and pits). Wall 3 resists lateral spread (ray anatomy). Wall 4 is the barrier zone [1,2,6]. The ranking is comparative, not a promise that Wall 1 always fails or that Wall 4 never fails. Boddy’s work on fungal community development inside standing trees showed that the living wood is a habitat in which succession continues after the first wall story has been told [3].

第 1 壁抵抗軸向擴展，在 Shigo 的排序中最弱。第 2 壁抵抗向內徑向擴展（晚材與紋孔）。第 3 壁抵抗側向擴展（射線解剖）。第 4 壁是屏障區 [1,2,6]。排序是比較性的，不是保證第 1 壁總失敗或第 4 壁永不失敗。Boddy 對站立木內真菌群落發育的研究顯示，活木材是一個棲地，在第一個壁故事說完之後，演替仍繼續 [3]。

The practical consequence is that a decay column's length, depth and tangential width are three different measurements. A long axial column with little radial depth is a different residual geometry from a short, deep cavity. Collapsing those three axes into a single 'percent decay' number discards the anatomy the walls describe.

操作後果是：腐朽柱的長度、深度與切向寬度是三次不同的測量。軸向長而徑向淺的柱，與短而深的空洞，是不同的剩餘幾何。把三軸壓成單一「腐朽百分比」，等於丟棄四壁所描述的解剖。

5. Cell scale: the chemistry the walls are made of

5. 細胞尺度：構成四壁的化學

At the finest scale the reaction zone is a deposit of phenolics, suberin and related compounds in cells that were alive or could still be metabolically modified at injury [4,7]. Tyloses and gums occlude vessels in many angiosperms; pit aspiration and resin canals do analogous work in many gymnosperms. These are not interchangeable tokens. A species that cannot form tyloses must spend a different account.

最細尺度上，反應區是酚類、木栓質及相關化合物在受傷時仍活或仍可被代謝修飾的細胞中的沉積 [4,7]。許多被子植物以侵填體與膠堵塞導管；許多裸子植物以紋孔封閉與樹脂道做類比工作。這些不是可互換代幣。不能形成侵填體的樹種必須花另一本賬。

The carbon and oxygen cost of that chemistry is not free. A tree already short of non-structural carbohydrate, or a root system already hypoxic in compacted soil, has less to spend on a new reaction front. Compartmentalization is therefore coupled to the carbon and hydraulic accounts treated in later physiological reviews; it is not a diagram that runs on empty.

該化學的碳與氧成本不是免費的。非結構性碳水化合物已短缺的樹，或在壓實土壤中已缺氧的根系，能花在新反應鋒上的資源更少。因此區隔化與後續生理綜述中的碳與水力賬戶耦合；它不是一張可以空轉的圖。

6. What compartmentalization does not decide

6. 區隔化所不能決定的事

A confined decay column can still remove enough section to change the failure mode of the remaining cylinder. Bond and Gruber, working from residual-wall geometry rather than from CODIT diagrams, argued that t/R thresholds treated as pass/fail laws are not supported as stand-alone criteria [8,9]. The wall story answers 'where did the fungus travel?'. It does not answer 'what load can the remaining wood carry in this weather envelope?' (Fig. 3).

被限制的腐朽柱仍可去掉足夠截面，從而改變剩餘圓柱的失效模式。Bond 與 Gruber 從剩餘壁幾何而不是從 CODIT 圖出發，主張把 t/R 閾值當成通過／失敗定律，並不作為獨立準則成立 [8,9]。壁的故事回答「真菌走到了哪」。它不回答「在此天氣包絡下，剩餘木材能承擔什麼荷載？」（圖 3）。

The operational sentence is therefore two-part: the species' compartmentalizing behaviour, and the residual geometry that behaviour has left. Either part used alone is an incomplete reading of the same stem.

操作句子因此是兩段：樹種的區隔行為，以及該行為留下的剩餘幾何。單獨使用任一段，都是對同一樹幹的不完整閱讀。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Trees respond to wood injury primarily by:

1. 樹木對木材傷害的主要回應是：

- A. Restoring the original tissue, as in animal healing.
A. 像動物癒合那樣恢復原組織。
- B. Walling off damaged tissue and growing new wood outside it.
B. 把受損組織封起來，並在其外長新木材。
- C. Raising P50 of the remaining xylem.
C. 提高剩餘木質部的 P50。
- D. Equalising Walls 1–4.
D. 使第 1–4 壁等強。

2. The reaction zone is formed in:

2. 反應區形成於：

- A. Only wood laid down after the injury.
A. 只有受傷後新造的木材。
- B. Wood that was already present at injury.
B. 受傷時已存在的木材。
- C. Only the inner bark.
C. 只有內皮。
- D. Only gymnosperm resin ducts.
D. 只有裸子植物樹脂道。

3. Wall 4 is strong relative to Walls 1–3 because it is:

3. 第 4 壁相對第 1–3 壁較強，是因為它是：

- A. The oldest heartwood.
A. 最老的心材。
- B. Anatomically new wood laid down after injury.
B. 受傷後新造、解剖上不同的木材。
- C. A paint film.
C. 一層漆膜。
- D. A t/R ratio.
D. 一個 t/R 比。

4. Shigo ranked which wall as the weakest against axial spread?

4. Shigo 把哪一壁排為抵抗軸向擴展最弱？

A. Wall 4.

A. 第 4 壁。

B. Wall 1.

B. 第 1 壁。

C. Wall 3.

C. 第 3 壁。

D. The bark.

D. 樹皮。

5. A successful barrier zone:

5. 成功的屏障區：

A. Guarantees mechanical safety of the stem.

A. 保證莖的力學安全。

B. Does not, by itself, certify residual load-bearing capacity.

B. 本身並不證明剩餘承載力。

C. Raises city canopy cover.

C. 提高城市樹冠覆蓋。

D. Equals a 10-20-30 rule.

D. 等於 10-20-30 規則。

6. Schwarze's comparative mycology implies that:

6. Schwarze 的比較真菌學意味：

A. All decay fungi breach reaction zones in the same way.

A. 所有腐朽菌以同一方式突破反應區。

B. White-rot, brown-rot and soft-rot use different strategies.

B. 白腐、褐腐與軟腐使用不同策略。

C. Fungi cannot enter living trees.

C. 真菌不能進入活樹。

D. CODIT applies only to palms.

D. CODIT 只適用於棕櫚。

7. Pearce treated the visible reaction-zone stain as:

7. Pearce 把可見的反應區色斑視為：

A. The defence itself.

A. 防禦本身。

B. A correlate of chemical deposits, not the defence token.

B. 化學沉積的相關物，不是防禦代幣。

C. Proof of zero residual risk.

C. 零剩餘風險的證明。

D. A legal register.

D. 法定名錄。

8. Axial length, radial depth and tangential width of a decay column:

8. 腐朽柱的軸向長度、徑向深度與切向寬度：

A. Are interchangeable percentages.

A. 是可互換的百分比。

B. Are three different measurements.

- B. 是三次不同的測量。
- C. Are defined only for palms.
- C. 只為棕櫚定義。
- D. Replace target assessment.
- D. 取代目標評估。

9. Bond and Gruber's geometric arguments imply that t/R:

9. Bond 與 Gruber 的幾何論點意味 t/R：

- A. Is a universal pass/fail law.
- A. 是普遍的通過／失敗定律。
- B. Is not a stand-alone failure criterion.
- B. 不是獨立的失效準則。
- C. Measures soil bulk density.
- C. 測量土壤容重。
- D. Equals Wall 4.
- D. 等於第 4 壁。

10. Compartmentalization chemistry is metabolically:

10. 區隔化化學在代謝上是：

- A. Free.
- A. 免費的。
- B. Costly, and coupled to carbon and oxygen supply.
- B. 昂貴的，並與碳與氧供應耦合。
- C. Paid by lightning conductors.
- C. 由避雷導體支付。
- D. Independent of the living tree.
- D. 獨立於活樹。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

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Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。